Department of Applied Mathematics and Statistics
COLORADO SCHOOL OF MINES
MATH 455: Partial Differential Equations

Assignment #4 - Heat Equation
Due Tuesday, October 20, 2026

1. (10 points) Let $k, L > 0$ and $f \in \mathrm{PC}^1[0, L]$ be a given function. Consider the heat equation

$$\frac{\partial^2 u}{\partial x^2} - \frac{1}{k}\frac{\partial u}{\partial t} = 0. \tag{HE}$$

(a) Write down the solution to (HE) with the boundary conditions

$$\frac{\partial u}{\partial x}(t, 0) = \frac{\partial u}{\partial x}(t, L) = 0, \qquad t > 0$$

and the initial condition

$$u(0, x) = f(x), \qquad 0 < x < L.$$

Solution — As with the wave equation, we apply an identical strategy of assuming the solution takes the separated form

$$u(t, x) = T(t)X(x)$$

Then

$$\frac{\partial^2 u}{\partial x^2} = T(t)X''(x) \qquad \frac{\partial u}{\partial t} = T'(t)X(x)$$

and plugging into the heat equation, we obtain:

$$k\frac{X''(x)}{X(x)} = \frac{T'(t)}{T(t)} = -\lambda$$

The left-hand side is a function of x only and the right-hand side is a function of t only. For them to be equal to one another for all values of t, x, they must both be equal to the same constant, which (for future convenience) we will notate as $-\lambda$. Hence our separated heat equation reduces to a system of independent, linear, homogeneous ordinary differential equations:

$$\begin{cases} kX''(x) + \lambda X(x) = 0 \\ T'(t) + \lambda T = 0 \end{cases}$$

We see that the structure of solutions to these equations depends on the sign of λ, so

we split into cases accordingly:

For $\lambda < 0$:

$$T(t) = c_0 e^{-\lambda t} \qquad X(x) = c_1 e^{\sqrt{-\frac{\lambda}{k}}x} + c_2 e^{-\sqrt{-\frac{\lambda}{k}}x}$$

For $\lambda = 0$:

$$T(t) = c_3 \qquad X(x) = c_4 + c_5 x$$

For $\lambda > 0$:

$$T(t) = c_6 e^{-\lambda t} \qquad X(x) = c_7 \cos\left(\sqrt{\frac{\lambda}{k}}x\right) + c_8 \sin\left(\sqrt{\frac{\lambda}{k}}x\right)$$

We now constrain these general solutions with the homogenous **Neumann** boundary conditions:

$$\frac{\partial u}{\partial x}(t,0) = 0 \implies T(t)X'(0) = 0 \implies X'(0) = 0$$
$$\frac{\partial u}{\partial x}(t,L) = 0 \implies T(t)X'(L) = 0 \implies X'(L) = 0$$

For $\lambda < 0$:

$$X'(0) = 0 \implies c_1\sqrt{-\frac{\lambda}{k}} - c_2\sqrt{-\frac{\lambda}{k}} = 0 \implies \boxed{c_1 = c_2}$$
$$X'(L) = 0 \implies c_1\sqrt{-\frac{\lambda}{k}}e^{\sqrt{-\frac{\lambda}{k}}L} - c_2\sqrt{-\frac{\lambda}{k}}e^{-\sqrt{-\frac{\lambda}{k}}L} = 0$$
$$\implies c_1\sqrt{-\frac{\lambda}{k}}\left(e^{\sqrt{-\frac{\lambda}{k}}L} - e^{-\sqrt{-\frac{\lambda}{k}}L}\right) = 0 \quad (\text{since } c_1 = c_2)$$
$$\implies \boxed{c_1 = 0} \quad (\text{Exponential factor} \neq 0 \text{ for } \lambda \neq 0)$$

For $\lambda = 0$:

$$X'(0) = X'(L) = 0 \implies \boxed{c_5 = 0}$$

For $\lambda > 0$:

$$
\begin{aligned}
X'(0) = 0 &\implies c_8\sqrt{\frac{\lambda}{k}} = 0 \\
&\implies \boxed{c_8 = 0} \quad (\text{since } \lambda > 0, k > 0) \\
X'(L) = 0 &\implies -c_7\sqrt{\frac{\lambda}{k}}\sin\left(\sqrt{\frac{\lambda}{k}}L\right) = 0 \\
&\implies \sin\left(\sqrt{\frac{\lambda}{k}}L\right) = 0 \quad (\text{since } \lambda \neq 0, k \neq 0, \text{ and assuming } c_7 \neq 0)
\end{aligned}
$$

Note that $c_7 = 0$ would produce a trivial solution since c_8 already equals zero, and the trivial solution is already accounted for in the other λ-cases.

$$
\begin{aligned}
&\implies \sqrt{\frac{\lambda}{k}}L = n\pi \quad (\text{for any } n \in \mathbb{Z}) \\
&\implies \boxed{\lambda = k\left(\frac{n\pi}{L}\right)^2}
\end{aligned}
$$

We write the general solutions produced by each case (with the Neumann boundary conditions applied):

For $\lambda < 0$:

$$
c_1 = c_2 = 0 \implies X(x) = 0 \implies u(t,x) = 0
$$

For $\lambda = 0$:

$$
c_5 = 0 \implies X(x) = c_4 \implies u(t,x) = c_3c_4 = C_0 \quad \text{for some } C_0 \in \mathbb{R}
$$

For $\lambda > 0$:

$$
c_8 = 0 \implies u_n(t,x) = C_n e^{-k\left(\frac{n\pi}{L}\right)^2 t}\cos\left(\frac{n\pi x}{L}\right) \quad \text{for } C_n \in \mathbb{R}
$$

Note that each separated eigenfunction solution will have its own constant C_n. Hence the general solution is the superposition of all solutions from each of these 3 cases:

$$
\boxed{u(t,x) = C_0 + \sum_{n=1}^{\infty} C_n e^{-k\left(\frac{n\pi}{L}\right)^2 t}\cos\left(\frac{n\pi x}{L}\right)}
$$

Let us now apply the initial condition $u(0,x) = f(x)$ for $0 \leq x \leq L$.

$$\implies C_0 + \sum_{n=1}^{\infty} C_n \cos\left(\frac{n\pi x}{L}\right) = f(x)$$

Notice that the application of this initial condition creates an equation that is exactly the Fourier series expression of $f(x)$. For both sides to be equal, we must have that each C_n is exactly the nth Fourier coefficient in the expansion of $f(x)$ on $[0, L]$. Hence

$$\boxed{C_0 = \frac{1}{L}\int_0^L f(x)dx}$$

$$\boxed{C_n = \frac{2}{L}\int_0^L f(x)\cos\left(\frac{n\pi x}{L}\right)dx \qquad n \geq 1}$$

(b) Let $T_1 > 0$, take

$$f(x) = T_1\frac{x}{L},$$

and determine the resulting solution by computing the Fourier coefficients and the corresponding Fourier series.

Solution —

$$\begin{aligned} C_0 &= \frac{1}{L}\int_0^L \left(T_1\frac{x}{L}\right)dx \\ &= \frac{T_1}{L^2}\int_0^L x dx \\ &= \frac{T_1}{L^2}\left(\frac{1}{2}L^2\right) \\ &= \frac{T_1}{2} \end{aligned}$$

For the general coefficients

$$\begin{aligned} C_n &= \frac{2}{L}\int_0^L \left(T_1\frac{x}{L}\right)\cos\left(\frac{n\pi x}{L}\right)dx \\ &= \frac{2T_1}{L^2}\int_0^L x\cos\left(\frac{n\pi x}{L}\right)dx \end{aligned}$$

Apply integration by parts:

$$\begin{aligned} U &= x & dV &= \cos\left(\frac{n\pi x}{L}\right)dx \\ dU &= dx & V &= \frac{L}{n\pi}\sin\left(\frac{n\pi x}{L}\right) \end{aligned}$$

$$
\begin{aligned}
C_n &= \frac{2T_1}{L^2}\left[\underbrace{\frac{Lx}{n\pi}\sin\left(\frac{n\pi x}{L}\right)\Big|_{x=0}^{x=L}}_{0} - \frac{L}{n\pi}\int_0^L \sin\left(\frac{n\pi x}{L}\right)dx\right] \\
&= \frac{-2T_1}{n\pi L}\left[-\frac{L}{n\pi}\cos\left(\frac{n\pi x}{L}\right)\right]_{x=0}^{x=L} \\
&= \frac{2T_1}{(n\pi)^2}\left[\cos\left(\frac{n\pi x}{L}\right)\right]_{x=0}^{x=L} \\
&= \frac{2T_1}{(n\pi)^2}\left[\cos(n\pi) - 1\right] \\
&= \frac{2T_1}{(n\pi)^2}\left[(-1)^n - 1\right]
\end{aligned}
$$

Hence the solution is

$$
\boxed{u(t,x) = \frac{T_1}{2} + \frac{2T_1}{\pi^2}\sum_{n=1}^{\infty}\left[\frac{(-1)^n - 1}{n^2}\right]e^{-k\left(\frac{n\pi}{L}\right)^2 t}\cos\left(\frac{n\pi x}{L}\right)}
$$

2. (15 points) Let $L, T_0 > 0$ and $f \in \mathrm{PC}^1[0, L]$ be given. Consider the temperature distribution of a long, thin bar in which heat is internally generated by a source with constant rate. In particular, we let $h > 0$ be given and consider the heat equation

$$
\frac{1}{k}\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2} + h
$$

for $0 < x < L$ and $t > 0$ with given boundary conditions

$$
u(t, 0) = T_0 \qquad \text{and} \qquad \frac{\partial u}{\partial x}(t, L) = 0
$$

and initial temperature distribution

$$
u(0, x) = f(x).
$$

Determine the temperature throughout the bar at all later times, namely compute $u(t, x)$.

Solution — Because this is the inhomogeneous heat equation (caused by the presence of the h term), we cannot separate the equation that results from assuming $u(t, x) = T(t)X(x)$ and plugging in. Hence we need a different method—namely, the one that involves finding a steady-state solution.

Assume the bar settles on the steady-state distribution $u_s(x)$ as $t \to \infty$. Because this

particular variation of the heat equation should be valid for all time, it's definitely valid for $u(t,x) = u_s(x)$ at $t = \infty$. Hence, plugging in $u_s(x)$, we obtain

$$\begin{aligned} 0 &= u_s''(x) + h \\ \implies u_s''(x) &= -h \\ \implies u_s'(x) &= -hx + C_0 \\ \implies u_s(x) &= -\frac{1}{2}hx^2 + C_0x + C_1 \end{aligned}$$

We can apply the boundary conditions to the steady-state solution:

$$\begin{aligned} u(t,0) = T_0 &\implies u_s(0) = T_0 \\ &\implies C_1 = T_0 \\ \frac{\partial u}{\partial x}(t,L) = 0 &\implies u_s'(L) = 0 \\ &\implies -hL + C_0 = 0 \\ &\implies C_0 = hL \end{aligned}$$

Hence the steady-state solution with applied boundary conditions is

$$u_s(x) = -\frac{1}{2}hx^2 + hLx + T_0$$

We can now perform a change-of-variable transformation that uses this steady-state solution to transform the given heat equation into a homogeneous (and separable) form. Namely, let $v(t,x)$ be the difference in heat between the current heat $u(t,x)$ and the eventual final heat $u_s(x)$:

$$v(t,x) = u(t,x) - u_s(x)$$

We plug this new expression of $u(t,x)$ into the given heat equation (Note that $u = v + u_s$)

$$\frac{1}{k}\frac{\partial v}{\partial t} = \frac{\partial^2 v}{\partial x^2} + u_s''(x) + h$$

And note that $u_s''(x) = -h$, so we recover the homogeneous form

$$\frac{1}{k}\frac{\partial v}{\partial t} = \frac{\partial^2 v}{\partial x^2}$$

With boundary conditions

$$\begin{aligned} u(t,0) = T_0 &\implies v(t,0) + \underbrace{u_s(0)}_{T_0} = T_0 \\ &\implies v(t,0) = 0 \end{aligned}$$

$$\frac{\partial u}{\partial x}(t,L) = 0 \implies \frac{\partial v}{\partial x}(t,L) + \underbrace{u_s'(L)}_{0} = 0$$
$$\implies \frac{\partial v}{\partial x}(t,L) = 0$$

And initial condition

$$u(0,x) = f(x) \implies v(0,x) + u_s(x) = f(x)$$
$$\implies v(0,x) = f(x) - u_s(x)$$

Since we've transformed the problem to a homogeneous form in $v(t,x)$, we proceed to solve regularly in v, assuming $v(t,x) = T(t)X(x)$. Then

$$k\frac{X''(x)}{X(x)} = \frac{T'(t)}{T(t)} = -\lambda$$

$$\begin{cases} kX''(x) + \lambda X(x) = 0 \\ T'(t) + \lambda T = 0 \end{cases}$$

The structures of solutions here are, again, dependent on the sign of λ, so we examine each one:

For $\lambda < 0$:

$$T(t) = c_0 e^{-\lambda t} \qquad X(x) = c_1 e^{\sqrt{-\frac{\lambda}{k}}x} + c_2 e^{-\sqrt{-\frac{\lambda}{k}}x}$$

For $\lambda = 0$:

$$T(t) = c_3 \qquad X(x) = c_4 + c_5 x$$

For $\lambda > 0$:

$$T(t) = c_6 e^{-\lambda t} \qquad X(x) = c_7 \cos\left(\sqrt{\frac{\lambda}{k}}x\right) + c_8 \sin\left(\sqrt{\frac{\lambda}{k}}x\right)$$

We apply the boundary conditions:

For $\lambda < 0$:

$$
\begin{aligned}
v(t,0) = 0 &\implies T(t)X(0) = 0 \\
&\implies X(0) = 0 \quad \text{(Assuming } T(t) \neq 0\text{)} \\
&\implies \boxed{c_1 = -c_2}
\end{aligned}
$$

$$
\begin{aligned}
&\frac{\partial v}{\partial x}(t,L) = 0 \\
&\quad\implies T(t)X'(L) = 0 \\
&\quad\implies X'(L) = 0 \quad \text{(Assuming } T(t) \neq 0\text{)} \\
&\quad\implies c_1\sqrt{-\frac{\lambda}{k}}e^{\sqrt{-\frac{\lambda}{k}}L} - c_2\sqrt{-\frac{\lambda}{k}}e^{-\sqrt{-\frac{\lambda}{k}}L} = 0 \\
&\quad\implies c_1\sqrt{-\frac{\lambda}{k}}\left(\underbrace{e^{\sqrt{-\frac{\lambda}{k}}L} + e^{-\sqrt{-\frac{\lambda}{k}}L}}_{\neq 0}\right) \\
&\quad\implies \boxed{c_1 = 0} \quad \text{(since } \lambda \neq 0, k \neq 0\text{)}
\end{aligned}
$$

For $\lambda = 0$:

$$
\begin{aligned}
X(0) = 0 &\implies \boxed{c_4 = 0} \\
X'(L) = 0 &\implies \boxed{c_5 = 0}
\end{aligned}
$$

For $\lambda > 0$:

$$
\begin{aligned}
X(0) = 0 &\implies \boxed{c_7 = 0} \\
X'(L) = 0 &\implies c_8\sqrt{\frac{\lambda}{k}}\cos\left(\sqrt{\frac{\lambda}{k}}L\right) = 0 \\
&\implies \sqrt{\frac{\lambda}{k}}L = \frac{\pi}{2} + n\pi \quad \text{for } n \in \mathbb{Z} \text{ (assuming } c_8 \neq 0\text{)} \\
&\implies \lambda = k\left(\frac{\pi}{2L} + \frac{n\pi}{L}\right)^2 \\
&\implies \lambda = \frac{k\pi^2}{L^2}\left(\frac{1}{2} + n\right)^2 \\
&\implies \lambda = \frac{k\pi^2}{4L^2}(2n+1)^2
\end{aligned}
$$

Our boundary condition-constrained constants cause $X(x) = 0$ and hence $u(t,x) = 0$ for $\lambda < 0$ and $\lambda = 0$. Hence, the superposition of all separated solutions consists only of $\lambda > 0$:

$$v(t,x) = \sum_{n=0}^{\infty} C_n e^{-\frac{k\pi^2(2n+1)^2}{4L^2}t} \sin\left(\frac{(2n+1)\pi x}{2L}\right)$$

We transform back to $u(t,x)$, using $u(t,x) = u_s(x) + v(t,x)$:

$$u(t,x) = -\frac{1}{2}hx^2 + hLx + T_0 + \sum_{n=0}^{\infty} C_n e^{-\frac{k\pi^2(2n+1)^2}{4L^2}t} \sin\left(\frac{(2n+1)\pi x}{2L}\right)$$

We now apply the initial condition

$$\begin{aligned} u(0,x) = f(x) &\implies -\frac{1}{2}hx^2 + hLx + T_0 + \sum_{n=0}^{\infty} C_n \sin\left(\frac{(2n+1)\pi x}{2L}\right) = f(x) \\ &\implies \sum_{n=0}^{\infty} C_n \sin\left(\frac{(2n+1)\pi x}{2L}\right) = f(x) + \frac{1}{2}hx^2 - hLx - T_0 \end{aligned}$$

We can't read off the C_n's as the coefficients of any identifiable Fourier series yet, because the summand isn't identical to the familiar $n\pi x/L$ frequency. Hence, we need to isolate the C_n's manually using the orthogonality of the sin eigenfunctions.

We multiply both sides by an equivalent m-frequency sin function and integrate from 0 to L. We'll start with the left-hand side:

$$\begin{aligned} &\int_0^L \sin\left(\frac{(2m+1)\pi x}{2L}\right) \sum_{n=0}^{\infty} C_n \sin\left(\frac{(2n+1)\pi x}{2L}\right) dx \\ &= \sum_{n=0}^{\infty} C_n \int_0^L \sin\left(\frac{(2m+1)\pi x}{2L}\right) \sin\left(\frac{(2n+1)\pi x}{2L}\right) dx \end{aligned}$$

Only the $n = m$ term in this sum is nonzero:

$$= C_m \int_0^L \sin\left(\frac{(2m+1)\pi x}{2L}\right)^2 dx$$

Let

$$u = \frac{(2m+1)\pi x}{2L} \qquad du = \frac{(2m+1)\pi}{2L} dx$$

The integral becomes

$$\begin{aligned}
&C_m \frac{2L}{(2m+1)\pi} \int_0^{\frac{(2m+1)\pi}{2}} \sin^2(u)du \\
&= C_m \frac{L}{(2m+1)\pi} \int_0^{\frac{(2m+1)\pi}{2}} 1-\cos(2u)du \quad \text{(applying sin power reduction)} \\
&= C_m \frac{L}{(2m+1)\pi} \left(\frac{(2m+1)\pi}{2}\right) - \frac{L}{(2m+1)\pi} \underbrace{\left[\frac{1}{2}\sin(2u)\right]_0^{\frac{(2m+1)\pi}{2}}}_{0} \\
&= C_m \frac{L}{2}
\end{aligned}$$

Hence, our 0-to-L-integrated, m-frequency-multiplied equation is

$$\begin{aligned}
C_m \frac{L}{2} &= \int_0^L \sin\left(\frac{(2m+1)\pi x}{2L}\right)\left(f(x)+\frac{1}{2}hx^2-hLx-T_0\right)dx \\
\implies C_m &= \frac{2}{L}\int_0^L \sin\left(\frac{(2m+1)\pi x}{2L}\right)\left(f(x)+\frac{1}{2}hx^2-hLx-T_0\right)dx
\end{aligned}$$

Hence the temperature of the bar at all later times is

$$\boxed{u(t,x) = -\frac{1}{2}hx^2 + hLx + T_0 + \sum_{n=0}^{\infty} C_n e^{-\frac{k\pi^2(2n+1)^2}{4L^2}t} \sin\left(\frac{(2n+1)\pi x}{2L}\right)}$$

With

$$\boxed{C_n = \frac{2}{L}\int_0^L \sin\left(\frac{(2n+1)\pi x}{2L}\right)\left(f(x)+\frac{1}{2}hx^2-hLx-T_0\right)dx}$$

3. (13 points) Let c be a given number satisfying $\frac{\pi}{10} < c < \frac{9\pi}{10}$ and define for every $0 < x < \pi$ the function

$$h(x) = \begin{cases} 5, & \text{if } c - \frac{\pi}{10} < x < c + \frac{\pi}{10} \\ 0, & \text{else.} \end{cases}$$

(a) Solve the heat equation with $k = 1$, namely

$$\frac{\partial^2 u}{\partial x^2} - \frac{\partial u}{\partial t} = 0$$

with

$$u(t,0) = u(t,\pi) = 0, \qquad t > 0$$

and

$$u(0,x) = h(x), \qquad 0 < x < \pi.$$

Use the trigonometric identity

$$\cos(\alpha) - \cos(\beta) = -2\sin\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha-\beta}{2}\right)$$

to simplify the form of the Fourier coefficients.

Solution — Since we already solved the general solution in problem 1, we solve the general heat equation identically (as before) and plug in $k = 1$ later. The separated general solutions $u(t,x) = T(t)X(x)$ to the homogenous heat equation takes the forms (as we've seen before in problem 1):

For $\lambda < 0$:

$$T(t) = c_0 e^{-\lambda t} \qquad X(x) = c_1 e^{\sqrt{-\frac{\lambda}{k}}x} + c_2 e^{-\sqrt{-\frac{\lambda}{k}}x}$$

For $\lambda = 0$:

$$T(t) = c_3 \qquad X(x) = c_4 + c_5 x$$

For $\lambda > 0$:

$$T(t) = c_6 e^{-\lambda t} \qquad X(x) = c_7 \cos\left(\sqrt{\frac{\lambda}{k}}x\right) + c_8 \sin\left(\sqrt{\frac{\lambda}{k}}x\right)$$

We now apply the **Dirichlet** boundary conditions:

$$\begin{aligned} u(t,0) = 0 &\implies T(t)X(0) = 0 \\ &\implies X(0) = 0 \quad (\text{for non-trivial } T(t) \neq 0) \\ u(t,\pi) = 0 &\implies T(t)X(\pi) = 0 \\ &\implies X(\pi) = 0 \quad (\text{for non-trivial } T(t) \neq 0) \end{aligned}$$

For $\lambda < 0$:

$$\begin{aligned} X(0) = 0 &\implies \boxed{c_1 = -c_2} \\ X(\pi) = 0 &\implies c_1 e^{\sqrt{-\frac{\lambda}{k}}\pi} + c_2 e^{-\sqrt{-\frac{\lambda}{k}}\pi} = 0 \\ &\implies c_1\left(e^{\sqrt{-\frac{\lambda}{k}}\pi} - e^{-\sqrt{-\frac{\lambda}{k}}\pi}\right) = 0 \quad (\text{using } c_1 = -c_2) \\ &\implies \boxed{c_1 = 0} \end{aligned}$$

(only the trivial solution exists in this case)

For $\lambda = 0$:

$$\begin{aligned} X(0) = 0 &\implies \boxed{c_4 = 0} \\ X(\pi) = 0 &\implies c_4 + \pi c_5 = 0 \\ &\implies \pi c_5 = 0 \\ &\implies \boxed{c_5 = 0} \end{aligned}$$

(only the trivial solution exists in this case)

For $\lambda > 0$:

$$\begin{aligned} X(0) = 0 &\implies \boxed{c_7 = 0} \\ X(\pi) = 0 &\implies c_8 \sin\left(\sqrt{\frac{\lambda}{k}}\pi\right) = 0 \\ &\implies \sqrt{\frac{\lambda}{k}}\pi = n\pi \quad \text{for } n \in \mathbb{Z} \text{ (and assuming } c_8 \neq 0) \\ &\implies \lambda = kn^2 \end{aligned}$$

Hence the general solution (the superposition of all our boundary-constrained solutions over all λ) is

$$\begin{aligned} u(t,x) &= \sum_{n=1}^{\infty} C_n e^{-kn^2 t} \sin(nx) \\ \text{or} \quad u(t,x) &= \sum_{n=1}^{\infty} C_n e^{-n^2 t} \sin(nx) \quad (\text{with } k = 1) \end{aligned}$$

We now apply the initial condition

$$\begin{aligned} & u(0,x) = h(x) \quad (\text{for } 0 \leq x \leq \pi) \\ \implies & \sum_{n=1}^{\infty} C_n \sin(nx) = h(x) \end{aligned}$$

This is exactly the Fourier sin expansion of $h(x)$ on $[0, \pi]$. Hence

$$C_n = \frac{2}{\pi}\int_0^{\pi} h(x)\sin(nx)dx$$

Note that

$$\frac{\pi}{10} < c < \frac{9\pi}{10} \implies c - \frac{\pi}{10} > 0 \text{ and } c + \frac{\pi}{10} < \pi$$

So the piecewise interval is contained fully in $[0, \pi]$:

$$\left[c - \frac{\pi}{10}, c + \frac{\pi}{10}\right] \in [0, \pi]$$

Hence

$$\begin{aligned} C_n &= \frac{2}{\pi}\int_{c-\frac{\pi}{10}}^{c+\frac{\pi}{10}} (5)\sin(nx)dx \\ &= \frac{10}{\pi}\int_{c-\frac{\pi}{10}}^{c+\frac{\pi}{10}} \sin(nx)dx \\ &= -\frac{10}{n\pi}\left[\cos(nx)\right]_{c-\frac{\pi}{10}}^{c+\frac{\pi}{10}} \\ &= -\frac{10}{n\pi}\left[\cos\left(n\left(c+\frac{\pi}{10}\right)\right) - \cos\left(n\left(c-\frac{\pi}{10}\right)\right)\right] \end{aligned}$$

We apply the identity

$$\cos(\alpha) - \cos(\beta) = -2\sin\left(\frac{\alpha+\beta}{2}\right)\sin\left(\frac{\alpha-\beta}{2}\right)$$

to get

$$C_n = \frac{20}{n\pi}\sin(nc)\sin\left(\frac{n\pi}{10}\right)$$

Hence the solution is

$$\boxed{u(t,x) = \frac{20}{\pi}\sum_{n=1}^{\infty}\frac{\sin(nc)}{n}\sin\left(\frac{n\pi}{10}\right)e^{-n^2t}\sin(nx)}$$

(b) Notice that for t sufficiently large, we have

$$u(t,x) \approx B_1 e^{-t}\sin(x).$$

What choice of c as defined above makes B_1 largest?

Solution — Note that for sufficiently large t, the upper solution modes (for $n = 2, 3, 4 \cdots$) become effectively zero in comparison to the $n = 1$ solution mode, due to their large(r) negative exponents. At least, the $n = 1$ solution mode decays the slowest, so the time-asymptotic behavior of the solution will, in the limit, consist of the first mode entirely. That is,

$$u(t,x) \sim \underbrace{\frac{20\sin(c)}{\pi}\sin\left(\frac{\pi}{10}\right)}_{B_1} e^{-t}\sin(x)$$

Maximize $B_1 \implies$ maximize $\sin(c) \implies$ get c as close to $\pi/2$ as possible. Since $\pi/2$ is already in the defined range of $\frac{\pi}{10} < c < \frac{9\pi}{10}$, we can just choose $c = \pi/2$.

(c) Using the idea in the previous part, answer the following question: Suppose that you're forced to put a cup of hot coffee somewhere between two blocks of ice. Where exactly should you put it if you want it to stay as warm as possible?

Solution — Note that c, as defined in the problem statement, is the center of the initial heat spike in the rod (the initial spike of temperature 5 exists from $x = c - \frac{\pi}{10}$ to $x = c + \frac{\pi}{10}$)

Also note that because of the homogeneous Dirichlet boundary conditions (forcing temperature zero at the ends of the rod), the two blocks of ice can be thought of as the ends of the rod. Correspondingly, the coffee can be thought of as the location of the c-centered temperature spike.

Hence answering this question reduces to answering the following: for what value of c will $u(t,x)$ be as large as possible, for as long as possible, near the location $x = c$?

Because we care only about the time-asymptotic behavior of u, we can assume $u(t,x) = B_1 e^{-t}\sin(x)$. Because $\sin(x)$ forms a positive hump on $[0,\pi]$, the temperature will always be largest in the middle, at $\pi/2$. To keep $u(t,x)$ large for as long as possible, we'll want its starting amplitude, B_1, to be as large as possible as well (which, conveniently, also happens at $c = \pi/2$).

Hence $c = \pi/2$ is the unambiguous best location to place the coffee to preserve its warmth. We want to place the coffee at the **midpoint** between the two blocks of ice.

4. (12 points) Let $f, g \in \mathrm{PC}^1[0,\pi]$ be given and find the solution of the inhomogeneous wave equation with speed of propagation $c = 1$, namely

$$\frac{\partial^2 u}{\partial t^2} - \frac{\partial^2 u}{\partial x^2} + 1 = 0$$

for $0 < x < \pi$ and $t > 0$ subject to the given boundary conditions

$$u(t,0) = 0 \qquad \text{and} \qquad u(t,\pi) = 0$$

for $t > 0$ and initial conditions

$$u(0,x) = f(x) \qquad \text{and} \qquad \frac{\partial u}{\partial t}(0,x) = g(x)$$

for $0 < x < \pi$.

Solution — Because this is the inhomogeneous wave equation, we must now follow a similar solution strategy to solving inhomogeneous differential equations: sum the homogeneous and particular solutions. We find the homogeneous solution first. Solve:

$$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = 0$$

As before, assuming the solution takes the form $u(t,x) = T(t)X(x)$ leads to the system of independent ODEs

$$\begin{cases} T'' + \lambda T = 0 \\ c^2 X'' + \lambda X = 0 \end{cases}$$

Which has several solutions depending on the sign of λ:

For $\lambda < 0$:

$$T(t) = c_0 e^{\sqrt{-\lambda}t} + c_1 e^{-\sqrt{-\lambda}t} \qquad X(x) = c_2 e^{\frac{1}{c}\sqrt{-\lambda}x} + c_3 e^{-\frac{1}{c}\sqrt{-\lambda}x}$$

For $\lambda = 0$:

$$T(t) = c_4 + c_5 t \qquad X(x) = c_6 + c_7 x$$

For $\lambda > 0$:

$$T(t) = c_8 \cos\left(\sqrt{\lambda}t\right) + c_9 \sin\left(\sqrt{\lambda}t\right) \qquad X(x) = c_{10} \cos\left(\frac{\sqrt{\lambda}}{c}x\right) + c_{11} \sin\left(\frac{\sqrt{\lambda}}{c}x\right)$$

We apply the (Dirichlet) boundary conditions:

$$\begin{aligned} u(t,0) = 0 &\implies T(t)X(0) = 0 \\ &\implies X(0) = 0 \quad \text{(For nontrivial } T(t) \neq 0\text{)} \\ u(t,\pi) = 0 &\implies T(t)X(\pi) = 0 \\ &\implies X(\pi) = 0 \quad \text{(For nontrivial } T(t) \neq 0\text{)} \end{aligned}$$

We constrain our unknown general constants by these boundary conditions:

For $\lambda < 0$:

$$\begin{aligned}
X(0) = 0 &\implies \boxed{c_2 = -c_3} \\
X(\pi) = 0 &\implies c_2 e^{\frac{1}{c}\sqrt{-\lambda}\pi} + c_3 e^{-\frac{1}{c}\sqrt{-\lambda}\pi} = 0 \\
X(\pi) = 0 &\implies c_2 \left(\underbrace{e^{\frac{1}{c}\sqrt{-\lambda}\pi} - e^{-\frac{1}{c}\sqrt{-\lambda}\pi}}_{\neq 0} \right) = 0 \quad (\text{using } -c_2 = c_3) \\
&\implies \boxed{c_2 = 0}
\end{aligned}$$

For $\lambda = 0$:

$$\begin{aligned}
X(0) = 0 &\implies \boxed{c_6 = 0} \\
X(\pi) = 0 &\implies c_5 + c_6\pi = 0 \\
&\implies \boxed{c_5 = 0}
\end{aligned}$$

For $\lambda > 0$:

$$\begin{aligned}
X(0) = 0 &\implies \boxed{c_{10} = 0} \\
X(\pi) = 0 &\implies c_{11} \sin\left(\frac{\sqrt{\lambda}}{c}\pi\right) = 0
\end{aligned}$$

We ignore $c_{11} = 0$ because, since $c_{10} = 0$ already, this would produce the trivial solution $X(x) = 0$ and hence $u(t, x) = 0$ that we already accounted for in previous λ cases. Hence

$$\begin{aligned}
&\implies \sin\left(\frac{\sqrt{\lambda}}{c}\pi\right) = 0 \\
&\implies \frac{\sqrt{\lambda}}{c}\pi = n\pi \quad (\text{for any } n \in \mathbb{Z}) \\
&\implies \lambda = c^2 n^2
\end{aligned}$$

Since the $\lambda < 0$ and $\lambda = 0$ cases don't contribute anything except the trivial $u(t, x) = 0$ solution, the general solution consists only of $\lambda > 0$ solutions (the superposition of all of them):

$$u_h(t, x) = \sum_{n=1}^{\infty} \Big[A_n \cos(cnt) + B_n \sin(cnt) \Big] \sin(nx)$$

Now we solve for the particular solution $u_p(t,x)$ to

$$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} + 1 = 0$$

In the spirit of the method of undetermined coefficients from ODEs, let's guess a particular solution that matches the form of the forcing function: -1. We guess $u(t,x) = A$ for some constant A, however we see that any constant A here would never satisfy the PDE due to both second derivatives being zero.

We guess $u(t,x) = f(x)$, some function independent of time, since the forcing function is also time-independent. Plugging into the PDE, we get

$$\begin{aligned} & c^2 f''(x) = 1 \\ \implies & f''(x) = \frac{1}{c^2} \\ \implies & f'(x) = \frac{1}{c^2}x + C_0 \\ \implies & f(x) = \frac{1}{2c^2}x^2 + C_0 x + C_1 \end{aligned}$$

We also constrain the particular solution by the same boundary conditions as we placed on the homogeneous solution:

$$\begin{aligned} u(t,0) = 0 \implies & f(0) = 0 \\ \implies & \boxed{C_1 = 0} \\ u(t,\pi) = 0 \implies & f(\pi) = 0 \\ \implies & \frac{\pi^2}{2c^2} + C_0 \pi = 0 \\ \implies & C_0 = -\frac{\pi}{2c^2} \end{aligned}$$

Hence the particular solution is

$$u_p(t,x) = \frac{1}{2c^2}x^2 - \frac{\pi}{2c^2}x$$

Once we find any particular solution, the general solution is thus the sum $u(t,x) = u_h(t,x) + u_p(t,x)$: (of course, with $c = 1$)

$$u(t,x) = \frac{1}{2c^2}x^2 - \frac{\pi}{2c^2}x + \sum_{n=1}^{\infty} \Big[A_n \cos(cnt) + B_n \sin(cnt) \Big] \sin(nx)$$

We now apply the initial conditions

$$\begin{aligned} u(0,x) = f(x) &\implies \frac{1}{2c^2}x^2 - \frac{\pi}{2c^2}x + \sum_{n=1}^{\infty} A_n \sin(nx) = f(x) \\ &\implies \sum_{n=1}^{\infty} A_n \sin(nx) = f(x) + \frac{\pi}{2c^2}x - \frac{1}{2c^2}x^2 \end{aligned}$$

Here we can read off the A_n coefficients as exactly those of the Fourier sin-expansion of the function

$$f(x) + \frac{\pi}{2c^2}x - \frac{1}{2c^2}x^2$$

on the interval $[0, \pi]$. Hence

$$\begin{aligned} A_n &= \frac{2}{\pi}\int_0^{\pi} \left(f(x) + \frac{\pi}{2c^2}x - \frac{1}{2c^2}x^2 \right) \sin(nx)dx \\ &= \frac{2}{\pi}\int_0^{\pi} f(x)\sin(nx)dx + \frac{1}{c^2}\int_0^{\pi} x\sin(nx)dx - \frac{1}{\pi c^2}\int_0^{\pi} x^2\sin(nx)dx \end{aligned}$$

Simplify the first integral:

$$\begin{aligned} \frac{1}{c^2}\int_0^{\pi} x\sin(nx)dx &= \frac{1}{c^2}\left[-\frac{x}{n}\cos(n\pi)\Big|_0^{\pi} + \frac{1}{n}\int_0^{\pi}\cos(nx)dx \right] \\ &= \frac{-\pi(-1)^n}{nc^2} \end{aligned}$$

Simplify the second integral:

$$\begin{aligned} \frac{1}{\pi c^2}\int_0^{\pi} x^2\sin(nx)dx &= \frac{1}{\pi c^2}\left[-\frac{x^2}{n}\cos(nx)\Big|_0^{\pi} + \frac{2}{n}\int_0^{\pi} x\cos(nx)dx \right] \\ &= \frac{1}{\pi c^2}\left[-\frac{\pi^2}{n}(-1)^n + \frac{2}{n}\left[\frac{x}{n}\sin(nx)\Big|_0^{\pi} - \frac{1}{n}\int_0^{\pi}\sin(nx)dx \right] \right] \\ &= \frac{1}{\pi c^2}\left[-\frac{\pi^2}{n}(-1)^n - \frac{2}{n^2}\int_0^{\pi}\sin(nx)dx \right] \\ &= \frac{1}{\pi c^2}\left[-\frac{\pi^2}{n}(-1)^n + \frac{2}{n^3}\left[\cos(nx)\right]_0^{\pi} \right] \\ &= \frac{1}{\pi c^2}\left[-\frac{\pi^2}{n}(-1)^n + \frac{2((-1)^n - 1)}{n^3} \right] \\ &= \frac{-\pi(-1)^n}{nc^2} + \frac{2((-1)^n - 1)}{\pi c^2 n^3} \end{aligned}$$

Hence

$$A_n = \frac{2}{\pi}\int_0^\pi f(x)\sin(nx)dx + \frac{2(1-(-1)^n)}{\pi c^2 n^3}$$

We now apply the second initial condition:

$$\begin{aligned}
\frac{\partial u}{\partial t}(0,x) = g(x) \implies & \left[\frac{\partial}{\partial t}\sum_{n=1}^{\infty}\Big[A_n\cos(cnt) + B_n\sin(cnt)\Big]\sin(nx)\right]_{(0,x)} = g(x) \\
& = \left[\sum_{n=1}^{\infty}\Big[-cnA_n\sin(cnt) + cnB_n\cos(cnt)\Big]\sin(nx)\right]_{(0,x)} \\
\implies & \sum_{n=1}^{\infty} cnB_n\sin(nx) = g(x) \\
\implies & \sum_{n=1}^{\infty} nB_n\sin(nx) = \frac{g(x)}{c}
\end{aligned}$$

Here we can read off nB_n as the Fourier coefficients of the sin-expansion of $g(x)/c$ on $[0,\pi]$:

$$\begin{aligned}
nB_n &= \frac{2}{\pi}\int_0^\pi \left(\frac{g(x)}{c}\right)\sin(nx)dx \\
\implies B_n &= \frac{2}{n\pi c}\int_0^\pi g(x)\sin(nx)dx
\end{aligned}$$

Hence, the solution to the PDE is (setting $c=1$) :

$$\boxed{u(t,x) = \frac{x^2-\pi x}{2} + \sum_{n=1}^{\infty}\Big[A_n\cos(nt) + B_n\sin(nt)\Big]\sin(nx)}$$

Where

$$\boxed{A_n = \frac{2}{\pi}\int_0^\pi f(x)\sin(nx)dx + \frac{2(1-(-1)^n)}{\pi n^3}}$$

$$\boxed{B_n = \frac{2}{n\pi}\int_0^\pi g(x)\sin(nx)dx}$$

Note that $f(x), g(x) \in PC^1[0,\pi]$, so these Fourier coefficients must exist.